Eggshell–Agar Biocomposites

Development and characterisation for sustainable interior applications

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Background

Material basis

94–97%

calcium carbonate by mass

0.85–2.66

specific gravity, below most mineral fillers

William & Owen (1995) · Torres et al. (2010) · Owuamanam & Cree (2020)

Two routes



The case for eggshell as a mineral phase is settled. This paper is about the binder.

Prior applications

Established uses of eggshell waste in engineered materials

Application	Binding phase	Reference
Partial cement replacement in concrete	Portland clinker	Teara et al. (2019); Bhartiya & Dubey (2018)
Compressed laterite brick	Soil – lime	Adogla et al. (2016)
Hot-mix asphalt	Bitumen	Razzaq & Yousif (2017)
Clay stabilisation	Clay mineral	Soundara & Vilasini (2015)
Eco-ceramic wall tile	Ceramic body	Vilarinho et al. (2022)
Particulate filler in PVC	Poly(vinyl chloride)	Skórczewska et al. (2020)
Filler in flexible polyurethane foam	Polyurethane	Victor et al. (2019)
Ceramic–metal composite	Metal matrix	Bilici et al. (2023); Yönetken et al. (2016)
Bone tissue scaffold	Calcium phosphate	Shafiei et al. (2019)
Wallpaper	Cellulosic furnish	Tutuş et al. (2018)

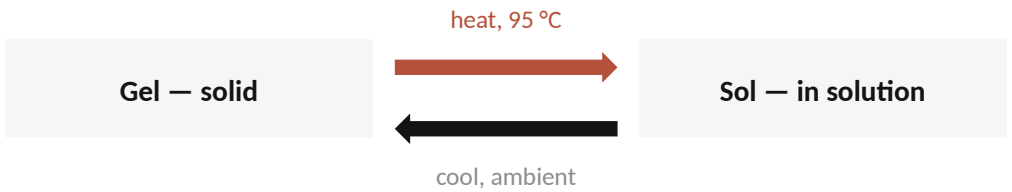
The burden is carried by the binder, and the binder is untouched.

MINERAL PHASE · recovered from waste in every study above

BINDING PHASE · cementitious, ceramic or petrochemical in every study above

Agar as binder

Thermoreversible setting



- No reactive hardener, no cure temperature
- 66% mineral loading, no coupling agent
- 32 glycerol formulations published as open data

Hernández et al. (2022), Materials 15(11), 3954

Form and loading case in this study

—	Film literature	This study
Form	Thin cast film	Cast solid, 20–50 mm
Mineral phase	Unfilled	66% of solids
Loading	Uniaxial tension	Compression and bending

The compositional logic transfers. The loading case does not, and for an interior surface that is bending.

Research gap

Three gaps follow from the preceding two slides

01	**The binding phase is never biological**	Both phases biological and biodegradable: not reported.
02	**Agar is characterised only as an unfilled film**	As the binder of a mineral-filled cast solid in bending: not reported.
03	**Property sets are reported in fragments**	All four properties for one formulation family: not reported.

It is not a novel constituent. It is a documented protocol and a single coherent property set.

Aim

To document a reproducible production protocol for a fully bio-based eggshell–agar composite, and to characterise its mechanical, reaction-to-fire and hygric behaviour against published test standards.

Objectives

01	02	03	04
Document the protocol	**Characterise to standards**	**Position against two families**	**Derive an evaluation criterion**

The fourth — deriving an evaluation criterion for this material class — is where I will spend most time.

Standards applied: ASTM D790-17 · ISO 11925-2:2020 · ASTM D570-98(2018).

Scope

Within scope

Dimension	This study
Formulations	Five — three compressive, two flexural
Specimens	Three per formulation for mechanical testing
Properties	Compression, flexure, ignitability, water absorption, bulk density
Standards	ASTM D790-17 · ISO 11925-2:2020 · ASTM D570-98(2018)
Conditions	Ambient cure, single laboratory, single operator
Application	Non-loadbearing interior surfaces

Not a specification

Characterisation data only.

Property coverage

Compression | **Flexure** | **Ignitability** | **Water absorption** | **Bulk density** | Durability | Fire classification | Life-cycle | Structural use

The filled boxes are what was determined. The outlined boxes are what was not.

Method

A material-driven sequence in four stages

1 → 2 → 3 → 4

Produce

Develop the composite and record every production parameter.

Measure

Characterise the cured material against published test standards.

Position

Benchmark the results against two material families.

Propose

Derive an evaluation criterion from the observed behaviour.

Properties first, application envelope second.

Material-driven design after Karana et al. (2015)

Research questions

01	**Which formulation variable governs compressive strength?**	Formulation
02	**How does plasticiser content affect flexural behaviour?**	Formulation
03	**How does the composite compare with conventional interior materials?**	Benchmark
04	**How does it compare with other bio-based composites?**	Benchmark
05	**Is compressive strength an adequate evaluation criterion for this class?**	Criterion

One to four are answered by measurement. Five is answered by argument.

Hypothesis

A fully bio-based eggshell–agar composite can be produced repeatably under ambient conditions, and its performance as an interior surface material is governed by deformation capacity rather than by compressive strength.

Two clauses, tested separately

Clause 1	Producible repeatably under ambient conditions	➡ tested on slides 11–14
Clause 2	Governed by deformation capacity rather than strength	➡ tested on slides 16–23

On provenance: the dissertation states a broader hypothesis framed on spatial resilience. This is its materials-engineering component, restated within the scope of these tests.

Stage 1 · Mineral phase

Progressive size reduction of the shell to a sieved powder



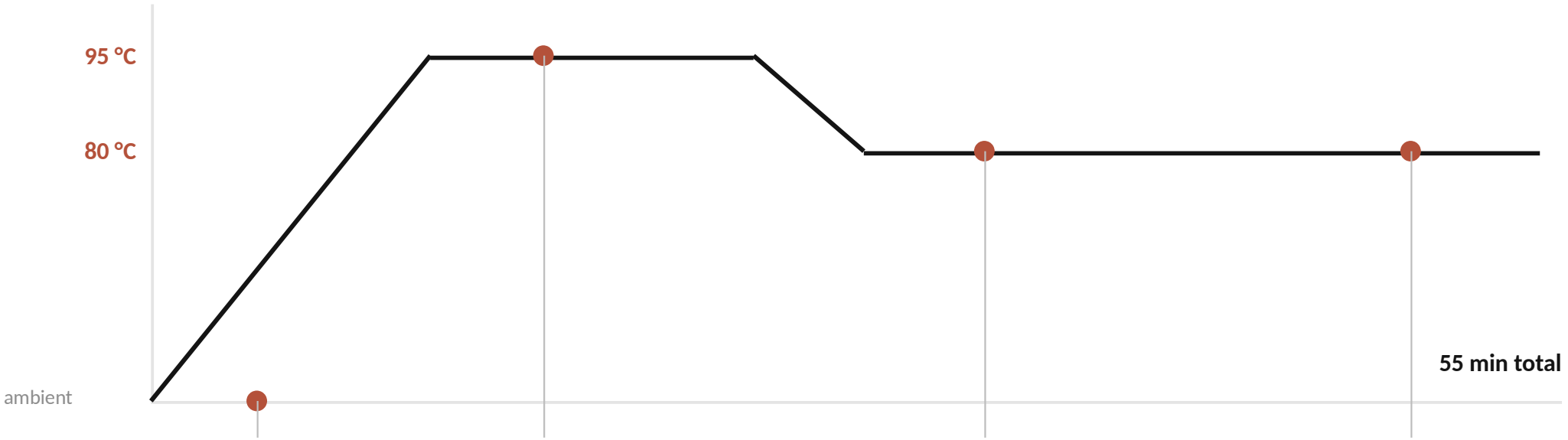
Whole shell	Cleaned	Heat-treated	Milled	Sieved
collected from a single bakery	membrane and residue removed manually	decontamination and drying	laboratory mechanical grinder	passing fraction retained
—	—	**120 °C, 20 min**	**to homogeneous powder**	**150 µm mesh**

That particle size was chosen to match the eggshell polymer-composite literature, so the mineral phase here is comparable with that work.

The 120 °C step is decontamination and drying, not sterilisation: sterility was not verified.

Stage 2 · Binder preparation

Temperature–time profile of the 55-minute preparation

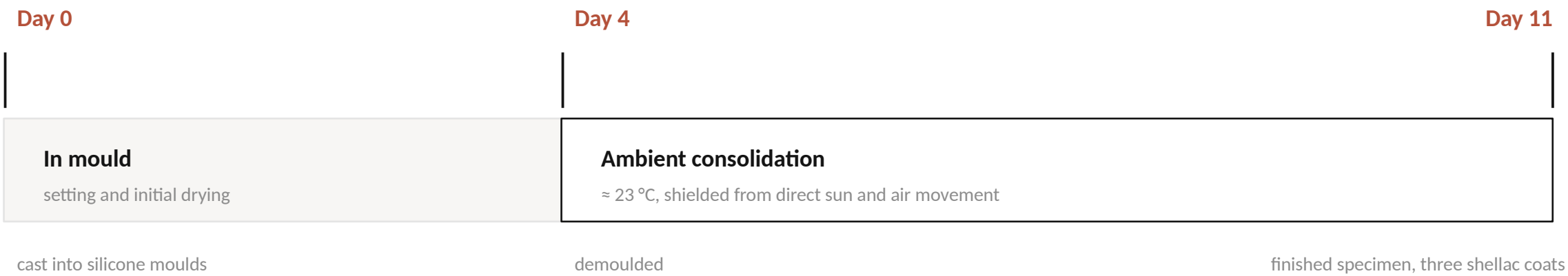


Agar is added incrementally under continuous stirring; bulk addition forms agglomerates that do not redisperse.

Schematic. The two set points and the total heating time are recorded; intermediate ramp rates are not. Stirring was manual.

Stage 3 · Casting and cure

Eleven days from casting to finished specimen, without thermal input



No thermal input after the 55-minute preparation: the entire cure is at room temperature.

Silicone moulds release without a parting agent and without damaging a partially consolidated specimen. 50 mm cube moulds for compression, flat moulds otherwise.

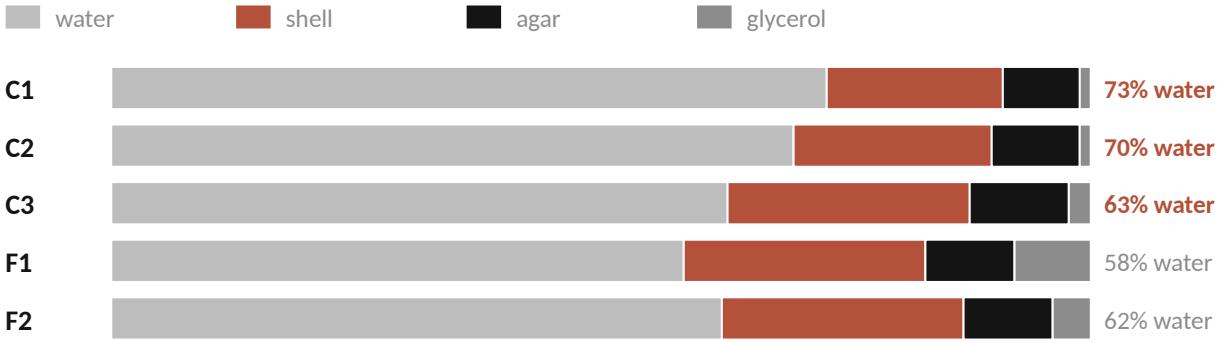
Limitation the composite shrinks as water evaporates, and post-drying dimensions were not logged, so every dimension I quote is a mould dimension.

Formulations

Five mixes, with the derived ratios that govern the results

ID	Test	Water g	Shell g	Agar g	Glycerol g	Shell : binder	Solids : water
C1	Compression	325	80	35	5	**2.00**	0.37
C2	Compression	310	90	40	5	**2.00**	0.44
C3	Compression	280	110	45	10	**2.00**	0.59
F1	Flexure	450	190	70	60	1.46	0.71
F2	Flexure	480	190	70	30	1.90	0.60

Batch composition by mass

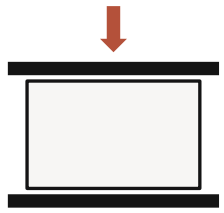


Solids concentration is the only variable — the water band shortens from C1 to C3.

Flexural series confounded: glycerol halved at fixed batch mass, so water rose with it.

Test programme

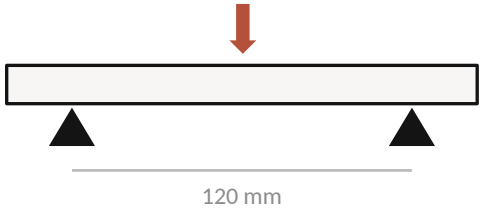
Four specimen geometries



Cube compression

50 mm cube, platen-loaded

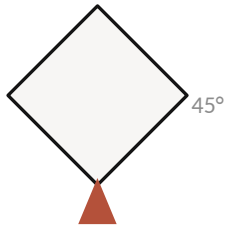
n = 3 per mix



Three-point flexure

180 × 60 × 20 mm, 120 mm span

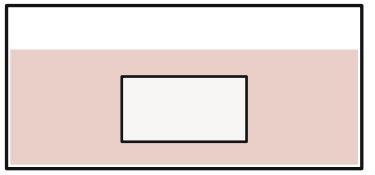
ASTM D790-17 · n = 3 per mix



Edge ignition

40 × 50 mm at 45°, 15 s flame

ISO 11925-2:2020 · n = 2



Immersion

40 × 50 mm, shellac-coated, 24 h

ASTM D570-98 · n = 1

Mechanical values are means of three specimens; individual values were not retained, so I cannot give standard deviations.

Ignitability, absorption and density rest on single determinations. · All testing: Advanced Construction Materials Research Laboratory, Hacettepe University. · Schematic, not to scale.

Compression

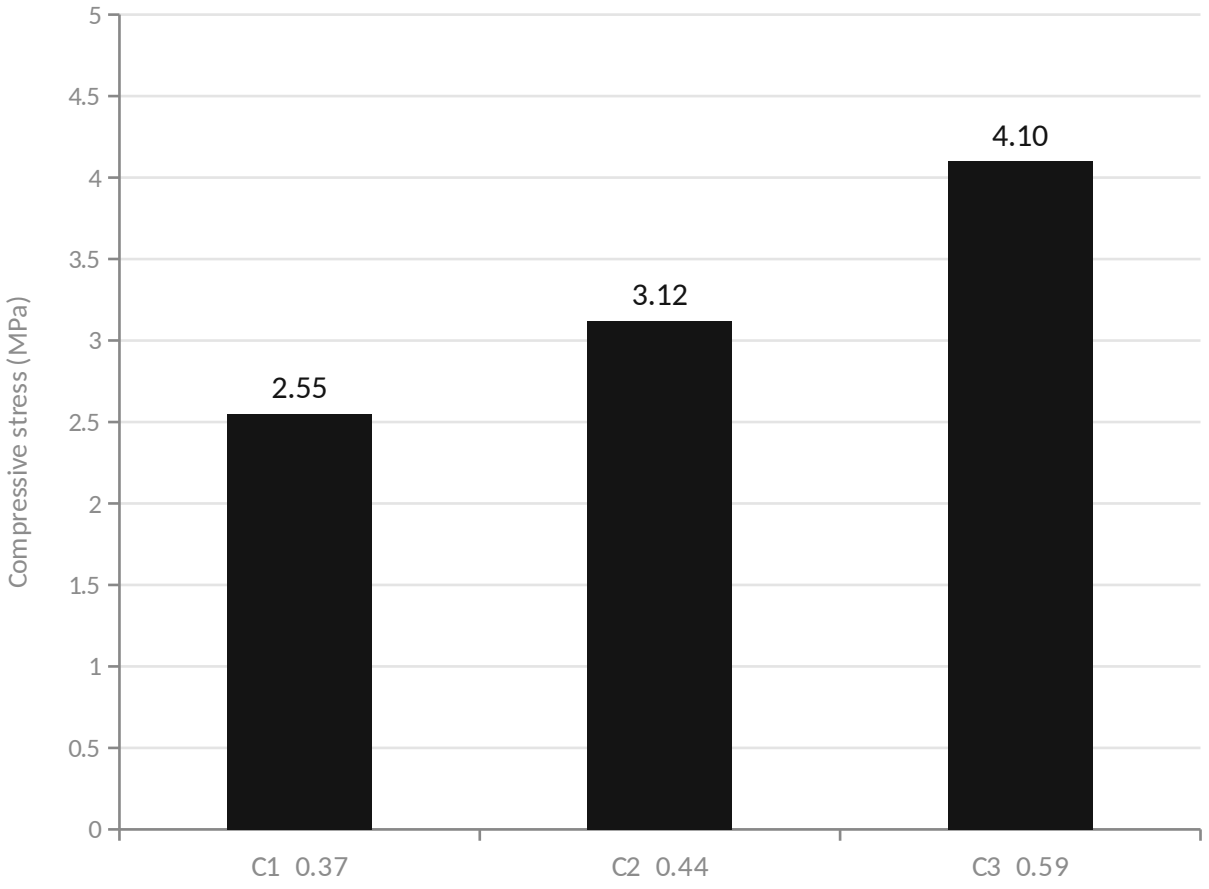
Measured values

ID	Solids : water	F_{max} kN	σc MPa
C1	0.37	3.89	2.55
C2	0.44	4.30	3.12
C3	0.59	**6.80**	**4.10**

Strength is governed by packing density, not by mineral content.

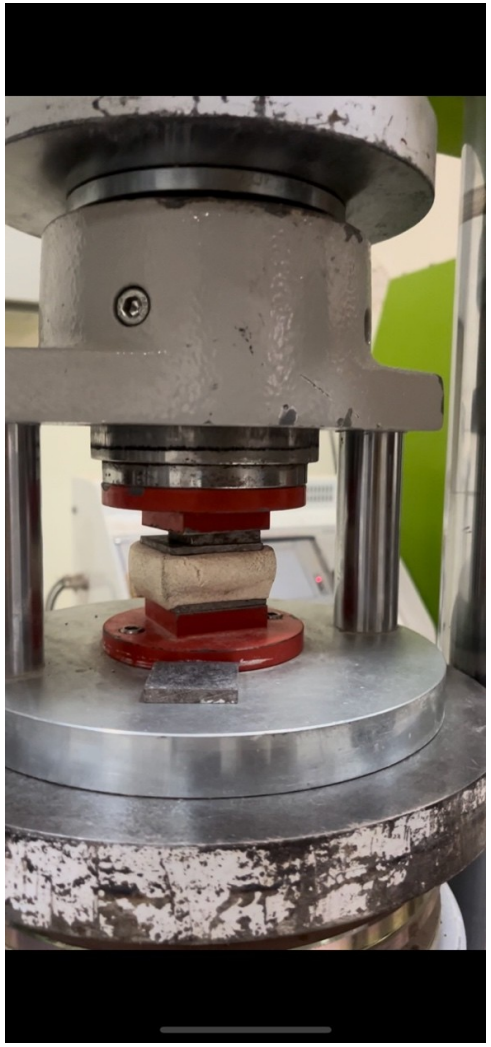
The route to a stronger composite is to remove water, not to add shell.

Linear over the range tested, $R^2 = 0.99$ — three formulations, therefore a hypothesis rather than a law. · Fitted relation: σc = 6.93 (solids : water) + 0.04

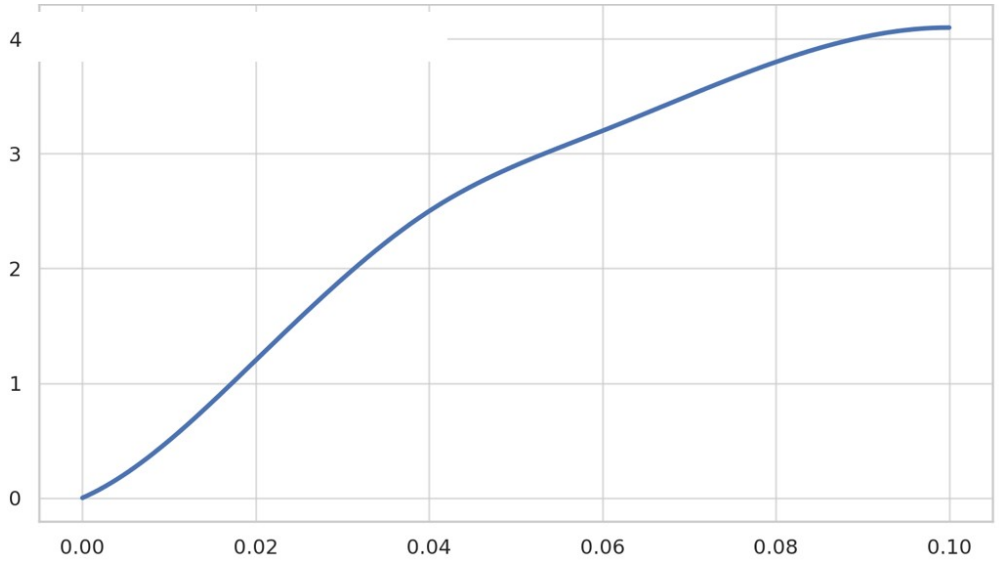


Solids : water ratio shown beneath each formulation identifier.

Compressive failure



Compressive stress (MPa) – strain, C3



Observed response

- Linear to approximately 1 MPa, then progressive loss of stiffness
- Peak stress near 10% strain, with no abrupt load drop
- Continued deformation beyond peak with retained integrity

Failure proceeds by progressive crushing rather than by fracture.

The cube has barrelled. It is shorter and wider, and still a single body.

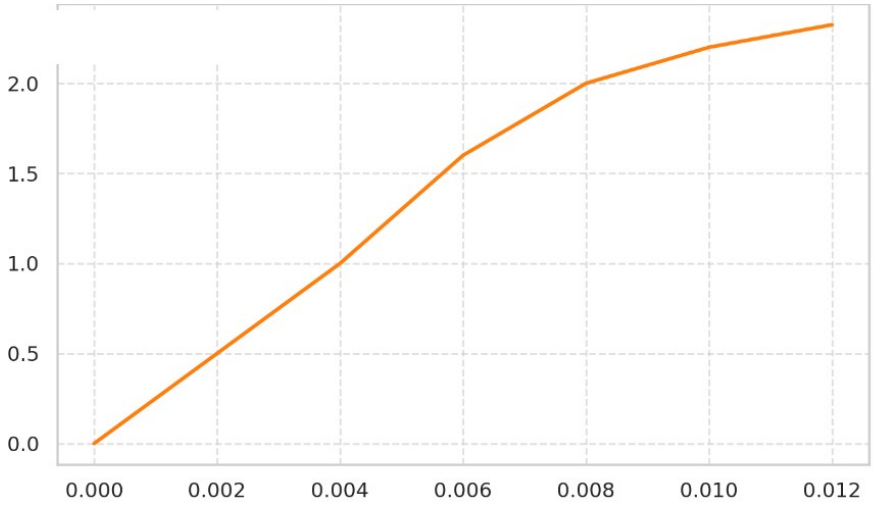
Strain derived from an assumed 10% maximum deformation based on observation, not instrumented measurement. · Bulk density 1.38 g/cm^3, mould-volume basis.

Flexure

Measured values — ASTM D790-17, three-point

ID	Glycerol % solids	F_{max} N	σF MPa	EF MPa	ε peak
F1	18.8	—	*not measurable*	—	—
F2	10.3	310	**2.325**	**192**	0.0121

Flexural stress (MPa) – strain, F2



The transition is sharp, and it lies between those two values.

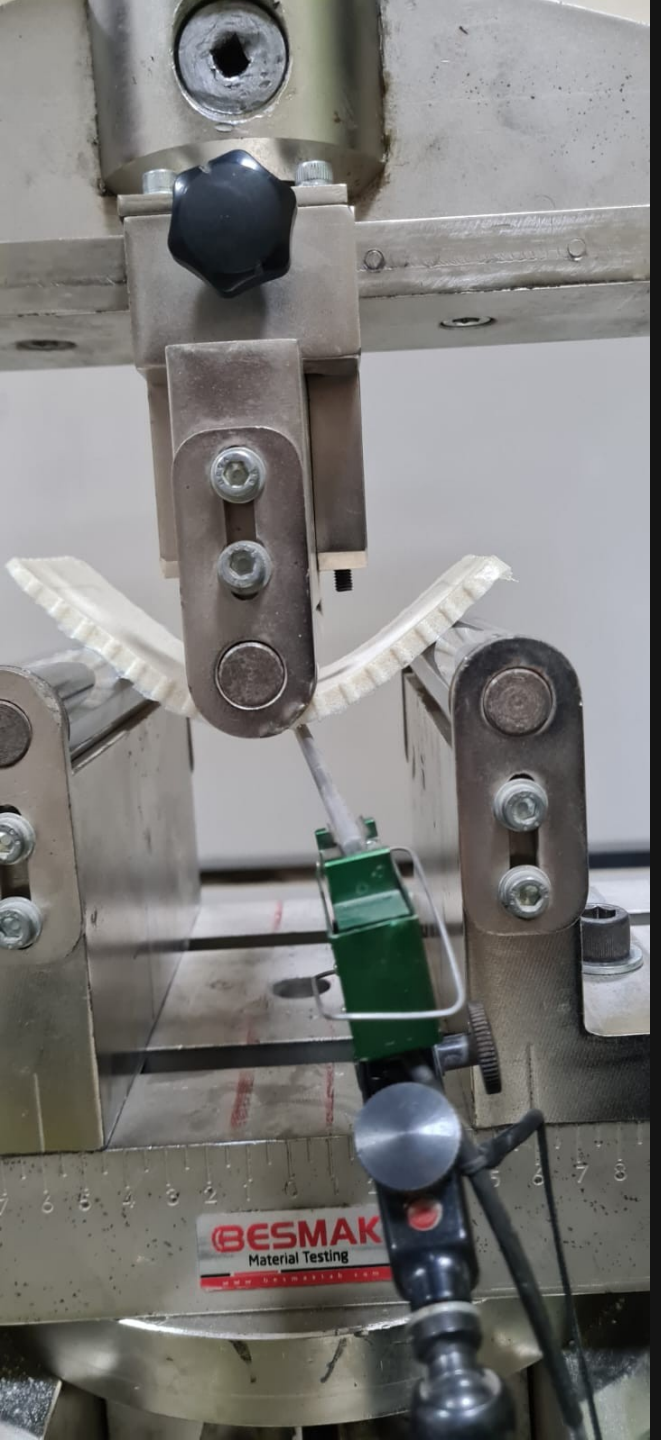
Flexural strength computed as $\sigma_F = 3FL / 2bd^2$ on the full rectangular section.

Limitation

The specimen is kerfed on its tension face. The standard assumes a solid rectangular section and the calculation uses the full section, so part of the measured compliance is geometric and the reported modulus is a lower bound.

Flexural resistance is governed by plasticiser content.

Confounded with solids concentration: batch mass was held constant, so water rose as glycerol fell.







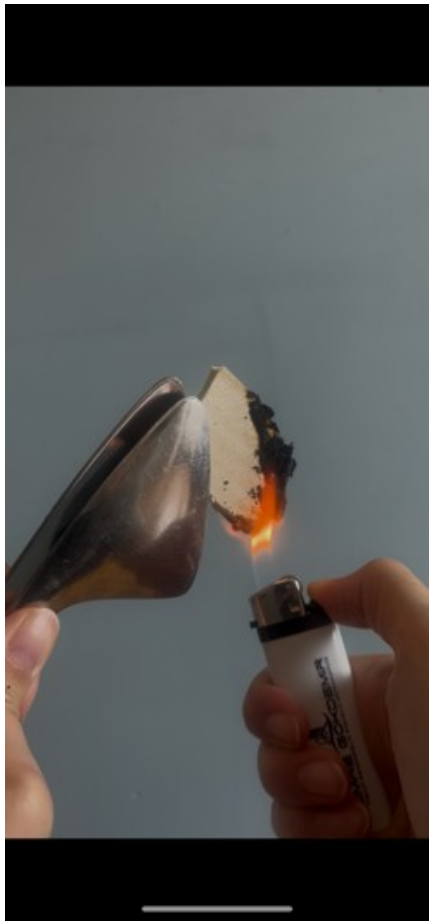
Flexural deformation

Both test programmes gave the same result: substantial deformation with retained integrity.

Deflected well beyond where a comparable ceramic parts. The kerfs are also why the reported modulus is a lower bound.

Three-point rig at high deflection · kerfed tension face · panel arched by hand

Fire, water, density



Ignitability

≈ 10 s

to ignition under direct flame

4 s

to self-extinguish after flame removal

- Charring remains local
- No lateral flame spread
- No melting and no dripping

This is an ignitability test; there is no Euroclass classification.

ISO 11925-2:2020. No cone calorimetry, smoke or toxicity data.

Water absorption, 24 h immersion

0.7%

mass gain, shellac-coated. Coating integrity is the limiting factor, not the composite.

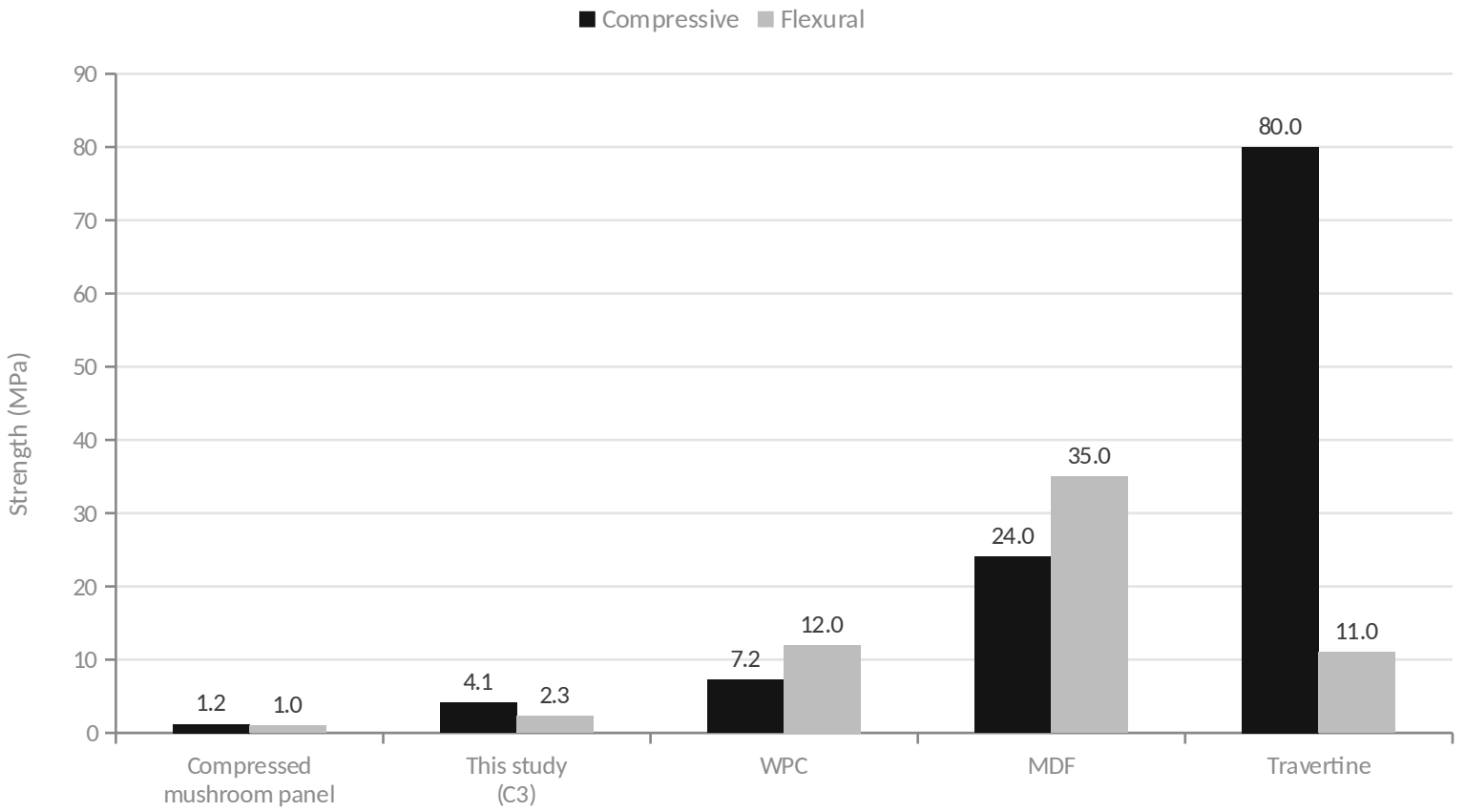
Bulk density

Material	g/cm³	Reference
This study (C3)	1.38	—
Mycelium, cellulose	0.3 – 0.8	Jones et al. (2020)
Fired clay brick	1.6 – 2.0	William & Owen (1995)

Mould-volume basis, therefore a lower bound.

Conventional materials

Compressive and flexural strength against interior surface materials



Velasco & Gaona (2011) · Klyosov (2007) · Nemli & Aydın (2007) · Altun (2006)

On strength alone this composite is not competitive with engineered wood or stone.

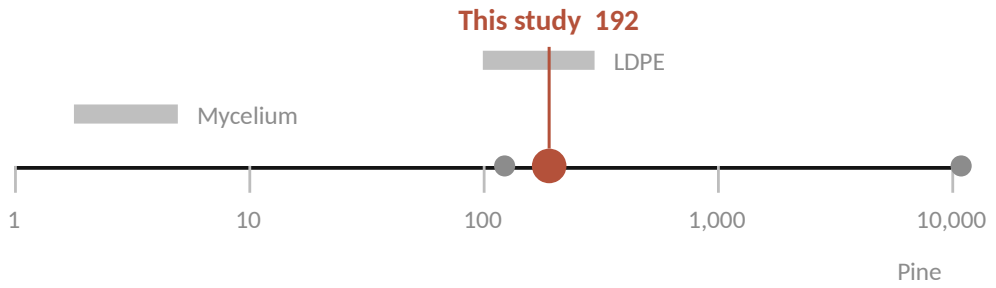
One sixth of MDF. One twentieth of travertine, a stone it visually resembles.

Bio-based composites

Compressive strength

Material	σc MPa	Reference
This study (C3)	**4.1**	—
Mycelium composite	0.145 – 0.452	Jones et al. (2020)
Hempcrete	0.3 – 0.6	Arnaud & Gourlay (2012)
Eggshell-filled PLA	2 – 5	Kim & Pal (2010)
Mineral-filled biocomposite	2.1 – 39.6	Shah et al. (2016)

Elastic modulus, MPa, logarithmic



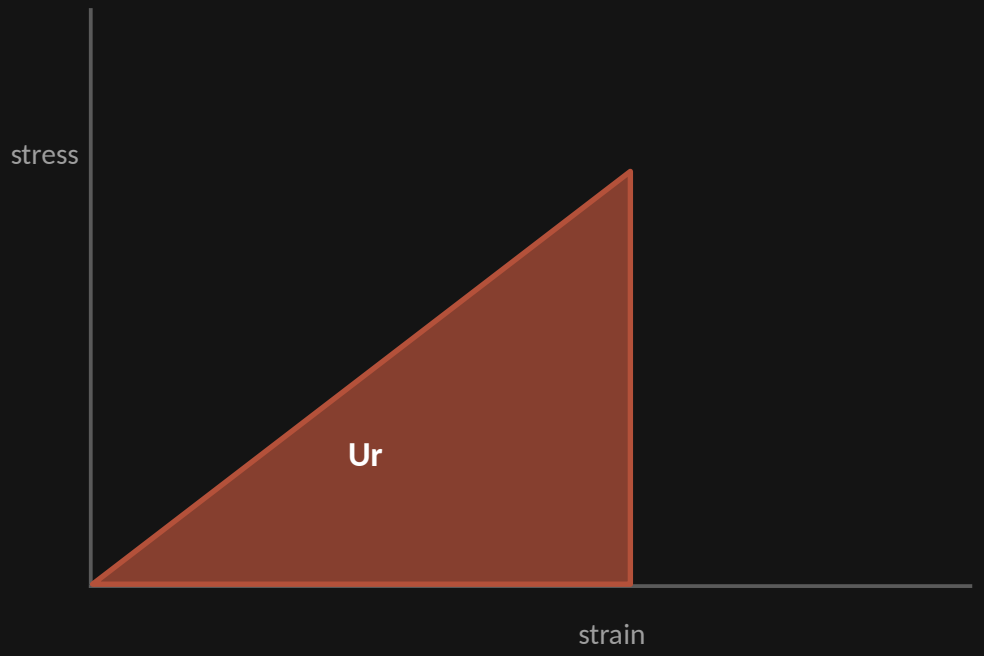
Unlabelled points: soy-based bioplastic 124, flax-fibre PLA 198, pine 11 000.

Water-cast, dried at room temperature, no oven, no autoclave, no extruder — and it reaches the stiffness of melt-processed bio-polymers.

An order of magnitude above the fungal and lime-binder family in compression; wood remains two orders of magnitude above. · Values are reported as published; specimen geometry and test method differ between sources.

Modulus of resilience

Elastic energy absorbed per unit volume



The shaded area is the energy absorbed per unit volume before permanent damage: ½ σ ε, and with ε = σ / E it reduces to the expression at right.

$$Ur = \sigma^2 / 2E$$

$$14.1\ kJ/m^3$$

computed from this study's own σ_F = 2.325 MPa and E_F = 192 MPa

Notice where the modulus sits: in the denominator.

Strength records the load at which a material stops performing, not whether it then fails gracefully or catastrophically.

Limitation

No comparative table is possible: the literature reports strength and modulus separately. A reporting convention, not a ranking.

Resilience as a property of systems that absorb disturbance and retain function, after Holling (1973).

Application envelope

Modular proposal derived from the measurements



Demountable modules on a reversible fixing, replaceable one unit at a time.

No life-cycle assessment was carried out, so these are qualitative potentials.

Conditions of use, each traced to a measured result

Non-loadbearing surfaces

self-weight and handling loads only; σc = 4.1 MPa

Dry interiors

not wet rooms, not façades; absorption depends on coating integrity

Away from ignition sources

self-extinguishing, but unclassified for reaction to fire

Modular rather than sheet

damage is replaced one unit at a time

Every constituent, the shellac included, is biodegradable; reversible fixing recovers the module intact. What is displaced is quarried limestone.

Limitations and further work

Each limitation, and the experiment that resolves it

Limitation	Resolved by
No dispersion data — individual specimen values not retained	**Repeat the series with individual values retained**
Compressive strain assumed from observation	**Instrument the strain; measure the curve**
Shrinkage unquantified, so density is a lower bound	**Record post-drying geometry**
Kerfed section calculated as a solid rectangle	**Test an un-kerfed section**
Flexural series confounded — glycerol and solids moved together	**Vary glycerol at constant solids concentration**
Durability, fire classification and life-cycle assessment untested	Outside this paper; named as further work

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Every value I reported today is a lower bound. I would rather set these out than have them found.

Conclusions

01	**A protocol that can be repeated**	every production parameter reported
02	**Packing density governs compressive strength**	not mineral content
03	**Plasticiser content sets brittle or ductile response**	the transition is sharp
04	**Bio-polymer stiffness without thermal processing**	water-cast, ambient-dried
05	**Strength alone is an insufficient criterion**	report σ and E together

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Testing: Advanced Construction Materials Research Laboratory, Hacettepe University, with thanks to Dr Hüseyin İlcan. Derived from the author's doctoral dissertation, Hacettepe University, 2025.

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